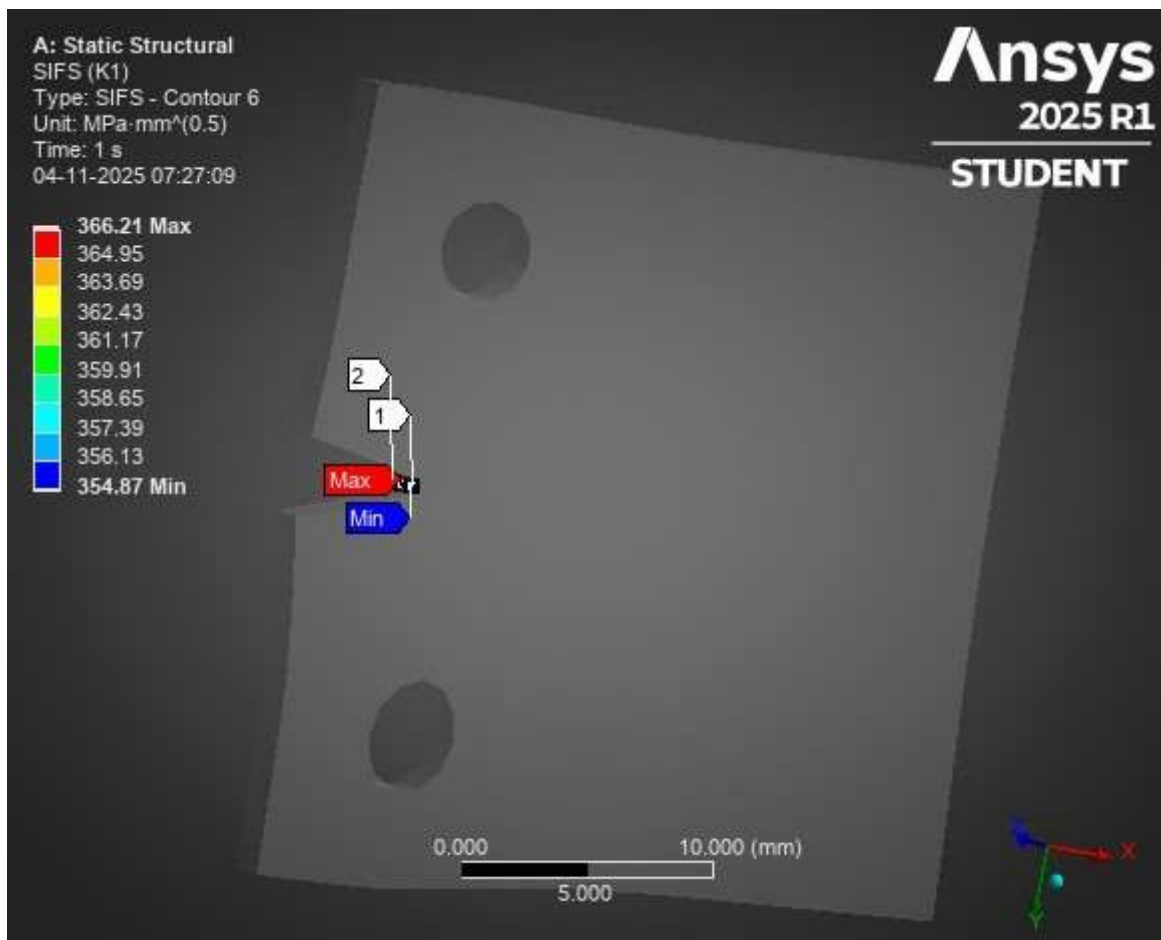


SIFS (K1)

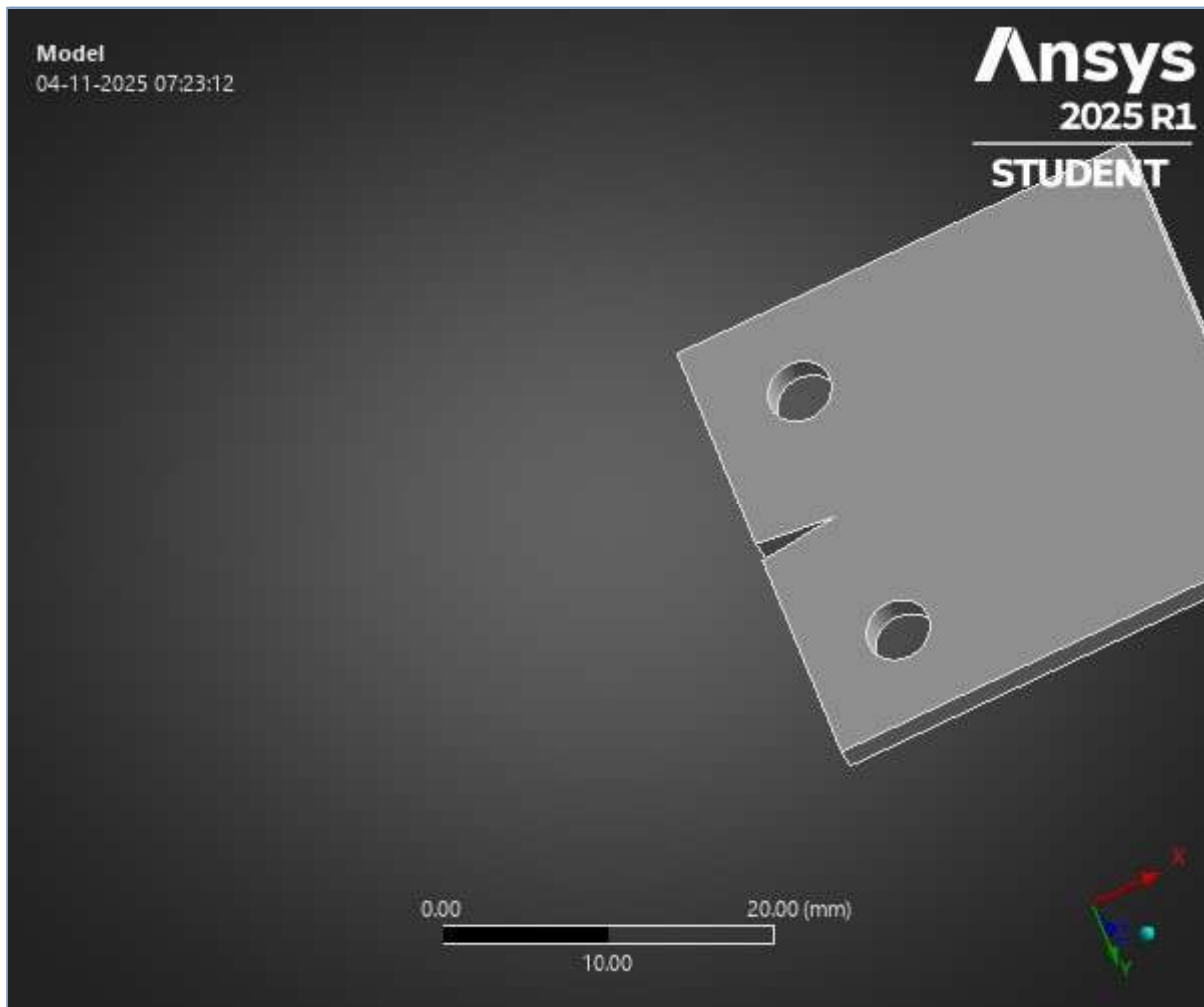
Subject: FRACTURE MECHANICS
Author: MOHANISH LANDE
Prepared For: PROJECT WORK
Date Wednesday, October 15, 2025
Comments:





Project

First Saved	Tuesday, October 14, 2025
Last Saved	Wednesday, October 15, 2025
Product Version	2025 R1
Save Project Before Solution	No
Save Project After Solution	No



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Units

TABLE 1

Unit System	Metric (mm, kg, N, s, mV, mA) Degrees rad/s Celsius
Angle	Degrees
Rotational Velocity	rad/s
Temperature	Celsius

Model (A4)

TABLE 2

Model (A4) > Geometry Imports

Object Name	<i>Geometry Imports</i>
State	Solved

TABLE 3

Model (A4) > Geometry Imports > Geometry Import (A3)

Object Name	<i>Geometry Import (A3)</i>
State	Solved
Definition	
Source	D:\fracture project final_files\dp0\SYS\DM\SYS.scdocx

Type	SpaceClaim
Basic Geometry Options	
Solid Bodies	Yes
Surface Bodies	Yes
Line Bodies	Yes
Parameters	Independent
Parameter Key	
Attributes	Yes
Attribute Key	
Named Selections	Yes
Named Selection Key	
Material Properties	Yes
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	Yes
Coordinate System Key	
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	No
Analysis Type	3-D
Mixed Import Resolution	None
Import Facet Quality	Source
Clean Bodies On Import	No
Stitch Surfaces On Import	None
Decompose Disjoint Geometry	Yes
Enclosure and Symmetry Processing	Yes

Geometry

TABLE 4
Model (A4) > Geometry

Object Name	Geometry
State	Fully Defined
Definition	
Source	D:\fracture project final_files\dp0\SYS\DM\SYS.scdocx
Type	SpaceClaim
Length Unit	Meters
Element Control	Program Controlled
Display Style	Body Color
Bounding Box	
Length X	30. mm
Length Y	30. mm
Length Z	2. mm
Properties	
Volume	1743.5 mm ³
Mass	1.3687e-002 kg
Scale Factor Value	1.
Statistics	
Bodies	1
Active Bodies	1
Nodes	15870

Elements	10436
Mesh Metric	None
Update Options	
Assign Default Material	No
Basic Geometry Options	
Solid Bodies	Yes
Surface Bodies	Yes
Line Bodies	Yes
Parameters	Independent
Parameter Key	
Attributes	Yes
Attribute Key	
Named Selections	Yes
Named Selection Key	
Material Properties	Yes
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	Yes
Coordinate System Key	
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	No
Analysis Type	3-D
Mixed Import Resolution	None
Import Facet Quality	Source
Clean Bodies On Import	No
Stitch Surfaces On Import	None
Decompose Disjoint Geometry	Yes
Enclosure and Symmetry Processing	Yes

TABLE 5
Model (A4) > Geometry > Parts

Object Name	SYS\Solid
State	Meshed
Graphics Properties	
Visible	Yes
Transparency	1
Definition	
Suppressed	No
Stiffness Behavior	Flexible
Coordinate System	Default Coordinate System
Reference Temperature	By Environment
Treatment	None
Material	
Assignment	Structural Steel
Nonlinear Effects	Yes
Thermal Strain Effects	Yes
Bounding Box	
Length X	30. mm
Length Y	30. mm
Length Z	2. mm

Properties	
Volume	1743.5 mm ³
Mass	1.3687e-002 kg
Centroid X	15.306 mm
Centroid Y	15. mm
Centroid Z	1. mm
Moment of Inertia Ip1	1.0321 kg·mm ²
Moment of Inertia Ip2	1.022 kg·mm ²
Moment of Inertia Ip3	2.045 kg·mm ²
Statistics	
Nodes	15870
Elements	10436
Mesh Metric	None
CAD Attributes	
PartTolerance:	0.00000001
SCRootPartComponent	
Color:143.175.143	

TABLE 6
Model (A4) > Materials

Object Name	<i>Materials</i>
State	Fully Defined
Statistics	
Materials	1
Material Assignments	0

Coordinate Systems

TABLE 7
Model (A4) > Coordinate Systems > Coordinate System

Object Name	Global Coordinate System	Coordinate System
State	Fully Defined	
Definition		
Type	Cartesian	
Coordinate System ID	0.	
Coordinate System		Program Controlled
APDL Name		
Suppressed		No
Origin		
Origin X	0. mm	5. mm
Origin Y	0. mm	15. mm
Origin Z	0. mm	1. mm
Define By		Geometry Selection
Geometry		Defined
Directional Vectors		
X Axis Data	[1. 0. 0.]	
Y Axis Data	[0. 1. 0.]	
Z Axis Data	[0. 0. 1.]	
Transfer Properties		
Source		
Read Only	No	
Principal Axis		

Axis		X
Define By		Global X Axis
Orientation About Principal Axis		
Axis		Y
Define By		Default
Transformations		
Base Configuration		Absolute
Transformed Configuration		[5. 15. 1.]

Mesh

TABLE 8
Model (A4) > Mesh

Object Name	<i>Mesh</i>
State	Solved
Display	
Display Style	Use Geometry Setting
Defaults	
Physics Preference	Mechanical
Element Order	Program Controlled
Element Size	Default (2.1237 mm)
Sizing	
Use Adaptive Sizing	No
Growth Rate	Default (1.85)
Max Size	2.1237e+005 mm
Mesh Defeaturing	Yes
Defeature Size	Default (1.0618e-002 mm)
Capture Curvature	No
Capture Proximity	No
Bounding Box Diagonal	42.474 mm
Average Surface Area	186.51 mm ²
Minimum Edge Length	2.0 mm
Quality	
Check Mesh Quality	Yes, Errors
Error Limits	Aggressive Mechanical
Target Element Quality	Default (5.e-002)
Smoothing	Medium
Mesh Metric	None
Inflation	
Use Automatic Inflation	None
Inflation Option	Smooth Transition
Transition Ratio	0.272
Maximum Layers	5
Growth Rate	1.2
Inflation Algorithm	Pre
Inflation Element Type	Wedges
View Advanced Options	No
Advanced	
Number of CPUs for Parallel Part Meshing	Program Controlled
Straight Sided Elements	No
Rigid Body Behavior	Dimensionally Reduced
Triangle Surface Mesher	Program Controlled

Topology Checking	Yes
Pinch Tolerance	Default (1.9113e-002 mm)
Generate Pinch on Refresh	No
Automatic Methods	
Sheet Body Method	Quad Dominant
Sweepable Body Method	Sweep
Statistics	
Nodes	15870
Elements	10436
Show Detailed Statistics	No

TABLE 9
Model (A4) > Mesh > Mesh Controls

Object Name	Edge Sizing	Patch Conforming Method
State	Fully Defined	
Scope		
Scoping Method	Geometry Selection	
Geometry	1 Edge	1 Body
Definition		
Suppressed	No	
Type	Sphere of Influence	
Sphere Center	Coordinate System	
Sphere Radius	1.5 mm	
Element Size	0.25 mm	
Control Messages	No	
Method		Tetrahedrons
Algorithm		Patch Conforming
Element Order		Use Global Setting
Advanced Improve Options		
Aggressive Thin Face Collapse		Program Controlled
Automatic Node Movement		Program Controlled
Refinement Options		
Refine at Thin Section		No

TABLE 10
Model (A4) > Fracture

Object Name	<i>Fracture</i>
State	Solved
Definition	
Re-mesh Hex-dominant to Tetrahedral	Off

TABLE 11
Model (A4) > Fracture > Pre-Meshed Crack

Object Name	<i>Pre-Meshed Crack</i>
State	Fully Defined
Scope	
Source	Pre-Meshed
Scoping Method	Named Selection
Crack Front (Named Selection)	Selection
Crack Faces Nodes	On
--Top Face Nodes	Selection 2
--Bottom Face Nodes	Selection 3
Definition	

Crack ID	40
Coordinate System	Coordinate System
Solution Contours	6
Symmetry	No
Suppressed	No

TABLE 12
Model (A4) > Fracture > SMART Crack Growth

Object Name	<i>SMART Crack Growth</i>
State	Fully Defined
Definition	
Analysis	Crack Growth
Method	SMART
Suppressed	No
Options for Crack Growth	
Initial Crack	Pre-Meshed Crack
Crack Growth Option	Static
Failure Criteria Option	Stress Intensity Factor
Critical Rate	1011.9 MPa·mm ^(0.5)
Min Increment of Crack Extension	Program Controlled
Max Increment of Crack Extension	Program Controlled
Stop Criterion	None
Remeshing Controls	
Mesh Coarsening	Conservative
Step Controls for Crack Growth	
Auto Time Stepping	Program Controlled
Initial Time Step	1.e-003 s
Minimum Time Step	1.e-004 s
Maximum Time Step	2.e-002 s

Named Selections

TABLE 13
Model (A4) > Named Selections > Named Selections

Object Name	<i>crack_tip</i>	<i>Selection</i>	<i>crack_top</i>	<i>crack_bottom</i>	<i>Selection 2</i>	<i>Selection 3</i>
State	Fully Defined					
Scope						
Scoping Method	Geometry Selection	Worksheet	Geometry Selection		Worksheet	
Geometry	1 Edge	17 Nodes	1 Face		258 Nodes	278 Nodes
Definition						
Send to Solver	Yes					
Protected	Program Controlled		Program Controlled			
Visible	Yes					
Program Controlled Inflation	Exclude					
Preserve During Solve		No			No	
Statistics						
Type	Manual					
Total Selection	1 Edge	17 Nodes	1 Face		258 Nodes	278 Nodes
Length	2. mm					

Suppressed	0		
Used by Mesh Worksheet	No		
Surface Area		10.077 mm ²	
Tolerance			
Tolerance Type		Program Controlled	Program Controlled
Zero Tolerance		1.e-008	1.e-008
Relative Tolerance		1.e-003	1.e-003
Angular Tolerance		1. °	1. °

Static Structural (A5)

TABLE 14
Model (A4) > Analysis

Object Name	<i>Static Structural (A5)</i>
State	Solved
Definition	
Physics Type	Structural
Analysis Type	Static Structural
Solver Target	Mechanical APDL
Options	
Environment Temperature	22. °C
Generate Input Only	No

TABLE 15
Model (A4) > Static Structural (A5) > Analysis Settings

Object Name	<i>Analysis Settings</i>
State	Fully Defined
Step Controls	
Number Of Steps	1.
Current Step Number	1.
Step End Time	1. s
Auto Time Stepping	Off
Define By	Substeps
Number Of Substeps	50.
Solver Controls	
Solver Type	Program Controlled
Weak Springs	Off
Solver Pivot Checking	Program Controlled
Large Deflection	Off
Inertia Relief	Off
Quasi-Static Solution	Off
Rotordynamics Controls	
Coriolis Effect	Off
Restart Controls	
Generate Restart Points	Program Controlled
Retain Files After Full Solve	No
Combine Restart Files	Program Controlled
Fracture Controls	
Fracture	On
SIFS	Yes

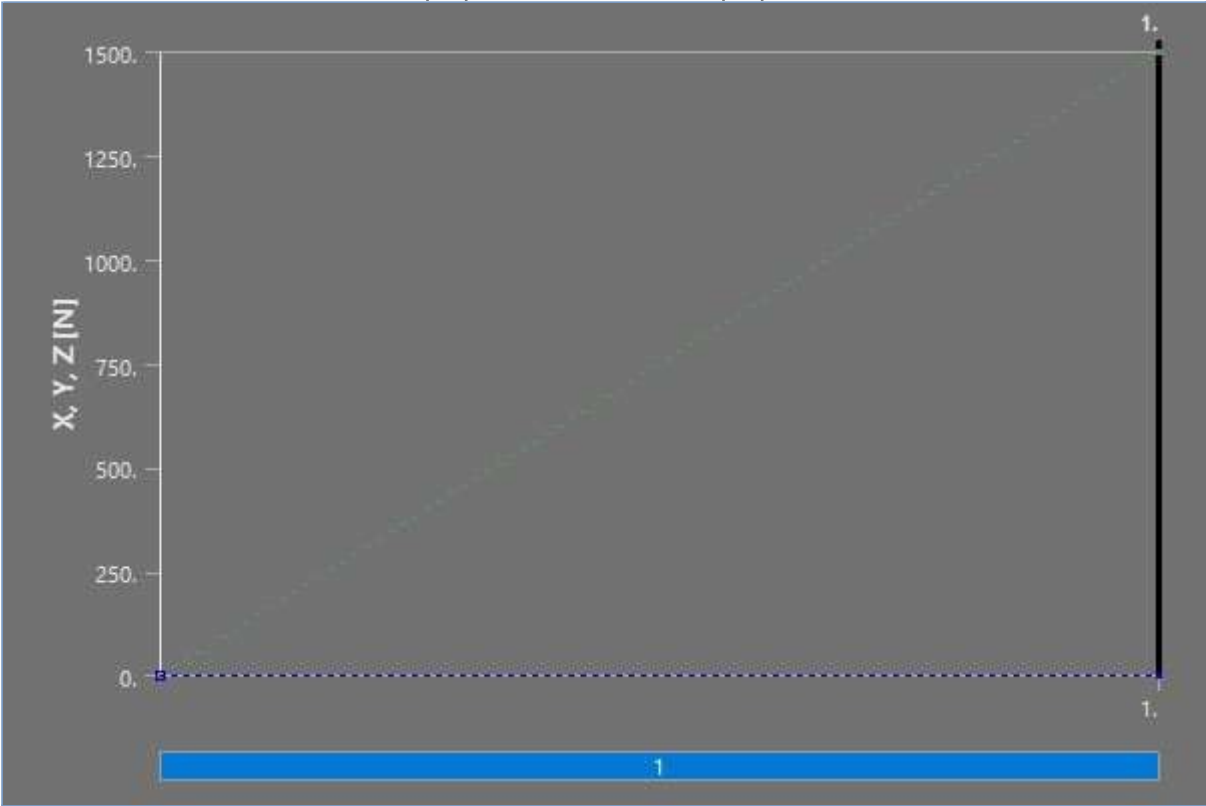
J-Integral	No
Material Force	No
T-Stress	No
Nonlinear Controls	
Newton-Raphson Option	Program Controlled
Force Convergence	Program Controlled
Moment Convergence	Program Controlled
Displacement Convergence	Program Controlled
Rotation Convergence	Program Controlled
Line Search	Program Controlled
Stabilization	Program Controlled
Advanced	
Inverse Option	No
Contact Split (DMP)	Program Controlled
Output Controls	
Output Selection	None
Stress	Yes
Back Stress	No
Strain	Yes
Contact Data	Yes
Nonlinear Data	No
Nodal Forces	No
Volume and Energy	Yes
Euler Angles	Yes
General Miscellaneous	No
Contact Miscellaneous	No
Store Results At	All Time Points
Result File Compression	Program Controlled
Analysis Data Management	
Solver Files Directory	D:\fracture project final_files\dp0\SYS\MECH\
Future Analysis	None
Scratch Solver Files Directory	
Save MAPDL db	No
Contact Summary	Program Controlled
Delete Unneeded Files	Yes
Nonlinear Solution	Yes
Solver Units	Active System
Solver Unit System	nmm

TABLE 16
Model (A4) > Static Structural (A5) > Loads

Model (A1) - Static Structural (A5) - Loads		
Object Name	Fixed Support	Force
State	Fully Defined	
Scope		
Scoping Method	Geometry Selection	
Geometry	1 Face	
Definition		
Type	Fixed Support	Force
Suppressed	No	
Define By		Components
Applied By		Surface Effect
Coordinate System		Global Coordinate System

X Component		0. N (ramped)
Y Component		1500. N (ramped)
Z Component		0. N (ramped)

FIGURE 1
Model (A4) > Static Structural (A5) > Force



Solution (A6)

TABLE 17
Model (A4) > Static Structural (A5) > Solution

Object Name	<i>Solution (A6)</i>
State	Solved
Adaptive Mesh Refinement	
Max Refinement Loops	1.
Refinement Depth	2.
Information	
Status	Done
MAPDL Elapsed Time	5 m 22 s
MAPDL Memory Used	786. MB
MAPDL Result File Size	200.31 MB
Post Processing	
Beam Section Results	No
On Demand Stress/Strain	No

TABLE 18
Model (A4) > Static Structural (A5) > Solution (A6) > Solution Information

Object Name	<i>Solution Information</i>
State	Solved

Solution Information	
Solution Output	Solver Output
Newton-Raphson Residuals	0
Identify Element Violations	0
Update Interval	2.5 s
Display Points	All
FE Connection Visibility	
Activate Visibility	Yes
Display	All FE Connectors
Draw Connections Attached To	All Nodes
Line Color	Connection Type
Visible on Results	No
Line Thickness	Single
Display Type	Lines

TABLE 19
Model (A4) > Static Structural (A5) > Solution (A6) > Results

Object Name	Directional Deformation	Equivalent Stress	Shear Stress
State	Solved		
Scope			
Scoping Method	Geometry Selection		
Geometry	All Bodies		
Definition			
Type	Directional Deformation	Equivalent (von-Mises) Stress	Shear Stress
Orientation	Y Axis		XY Component
By	Time		
Display Time	Last		
Separate Data by Entity	No		
Coordinate System	Global Coordinate System		Global Coordinate System
Calculate Time History	Yes		
Identifier			
Suppressed	No		
Results			
Minimum	-3.5565e-003 mm	7.5347e-002 MPa	-168.81 MPa
Maximum	1.2465e-002 mm	481.68 MPa	182.43 MPa
Average	4.0025e-003 mm	104.55 MPa	-2.2454 MPa
Minimum Occurs On	SYS\Solid		
Maximum Occurs On	SYS\Solid		
Minimum Value Over Time			
Minimum	-3.5565e-003 mm	1.5069e-003 MPa	-168.81 MPa
Maximum	-7.1129e-005 mm	7.5347e-002 MPa	-3.3761 MPa
Maximum Value Over Time			
Minimum	2.493e-004 mm	9.6337 MPa	3.6485 MPa
Maximum	1.2465e-002 mm	481.68 MPa	182.43 MPa
Information			
Time	1. s		
Load Step	1		
Substep	50		
Iteration Number	50		
Integration Point Results			
Display Option		Averaged	
Average Across Bodies		No	

FIGURE 2
Model (A4) > Static Structural (A5) > Solution (A6) > Directional Deformation

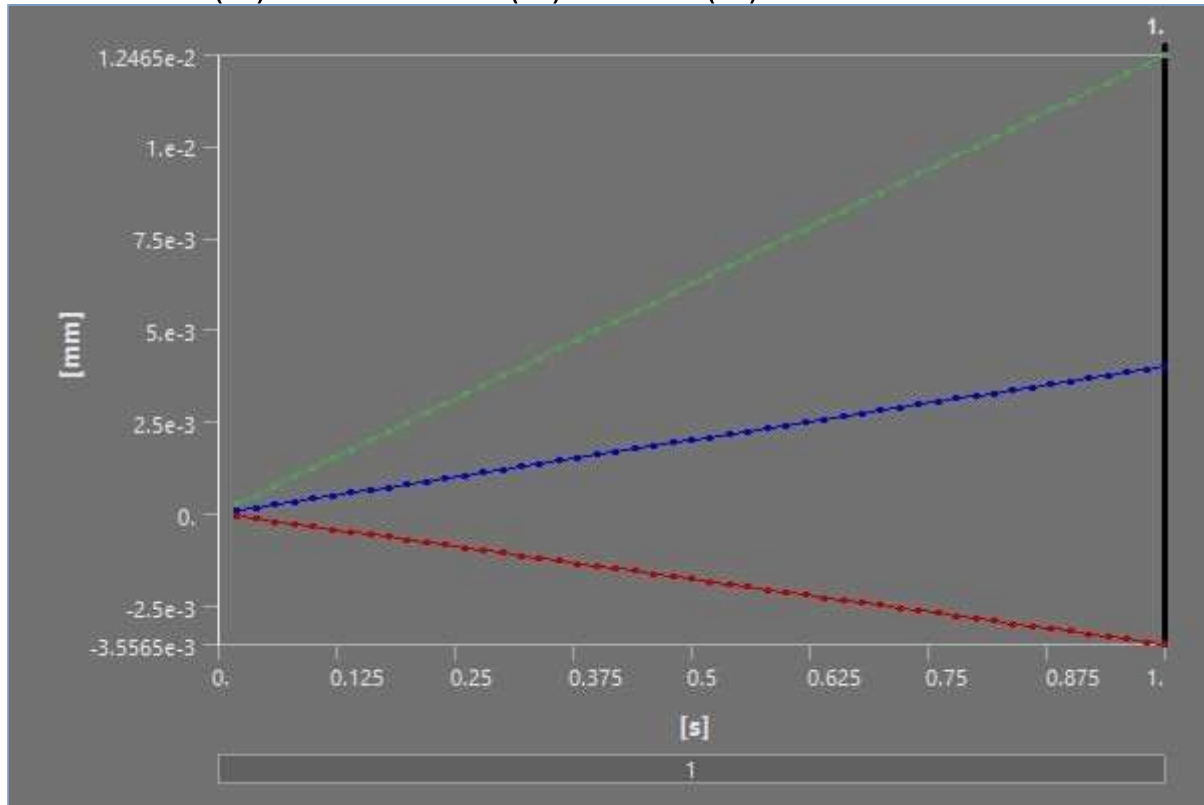


TABLE 20
Model (A4) > Static Structural (A5) > Solution (A6) > Directional Deformation

Time [s]	Minimum [mm]	Maximum [mm]	Average [mm]
2.e-002	-7.1129e-005	2.493e-004	8.0049e-005
4.e-002	-1.4226e-004	4.9861e-004	1.601e-004
6.e-002	-2.1339e-004	7.4791e-004	2.4015e-004
8.e-002	-2.8452e-004	9.9722e-004	3.202e-004
0.1	-3.5565e-004	1.2465e-003	4.0025e-004
0.12	-4.2678e-004	1.4958e-003	4.8029e-004
0.14	-4.979e-004	1.7451e-003	5.6034e-004
0.16	-5.6903e-004	1.9944e-003	6.4039e-004
0.18	-6.4016e-004	2.2437e-003	7.2044e-004
0.2	-7.1129e-004	2.493e-003	8.0049e-004
0.22	-7.8242e-004	2.7424e-003	8.8054e-004
0.24	-8.5355e-004	2.9917e-003	9.6059e-004
0.26	-9.2468e-004	3.241e-003	1.0406e-003
0.28	-9.9581e-004	3.4903e-003	1.1207e-003
0.3	-1.0669e-003	3.7396e-003	1.2007e-003
0.32	-1.1381e-003	3.9889e-003	1.2808e-003
0.34	-1.2092e-003	4.2382e-003	1.3608e-003
0.36	-1.2803e-003	4.4875e-003	1.4409e-003
0.38	-1.3515e-003	4.7368e-003	1.5209e-003
0.4	-1.4226e-003	4.9861e-003	1.601e-003
0.42	-1.4937e-003	5.2354e-003	1.681e-003
0.44	-1.5648e-003	5.4847e-003	1.7611e-003
0.46	-1.636e-003	5.734e-003	1.8411e-003

0.48	-1.7071e-003	5.9833e-003	1.9212e-003
0.5	-1.7782e-003	6.2326e-003	2.0012e-003
0.52	-1.8494e-003	6.4819e-003	2.0813e-003
0.54	-1.9205e-003	6.7312e-003	2.1613e-003
0.56	-1.9916e-003	6.9805e-003	2.2414e-003
0.58	-2.0627e-003	7.2298e-003	2.3214e-003
0.6	-2.1339e-003	7.4791e-003	2.4015e-003
0.62	-2.205e-003	7.7285e-003	2.4815e-003
0.64	-2.2761e-003	7.9778e-003	2.5616e-003
0.66	-2.3473e-003	8.2271e-003	2.6416e-003
0.68	-2.4184e-003	8.4764e-003	2.7217e-003
0.7	-2.4895e-003	8.7257e-003	2.8017e-003
0.72	-2.5607e-003	8.975e-003	2.8818e-003
0.74	-2.6318e-003	9.2243e-003	2.9618e-003
0.76	-2.7029e-003	9.4736e-003	3.0419e-003
0.78	-2.774e-003	9.7229e-003	3.1219e-003
0.8	-2.8452e-003	9.9722e-003	3.202e-003
0.82	-2.9163e-003	1.0222e-002	3.282e-003
0.84	-2.9874e-003	1.0471e-002	3.3621e-003
0.86	-3.0586e-003	1.072e-002	3.4421e-003
0.88	-3.1297e-003	1.0969e-002	3.5222e-003
0.9	-3.2008e-003	1.1219e-002	3.6022e-003
0.92	-3.2719e-003	1.1468e-002	3.6823e-003
0.94	-3.3431e-003	1.1717e-002	3.7623e-003
0.96	-3.4142e-003	1.1967e-002	3.8424e-003
0.98	-3.4853e-003	1.2216e-002	3.9224e-003
1.	-3.5565e-003	1.2465e-002	4.0025e-003

FIGURE 3
Model (A4) > Static Structural (A5) > Solution (A6) > Equivalent Stress

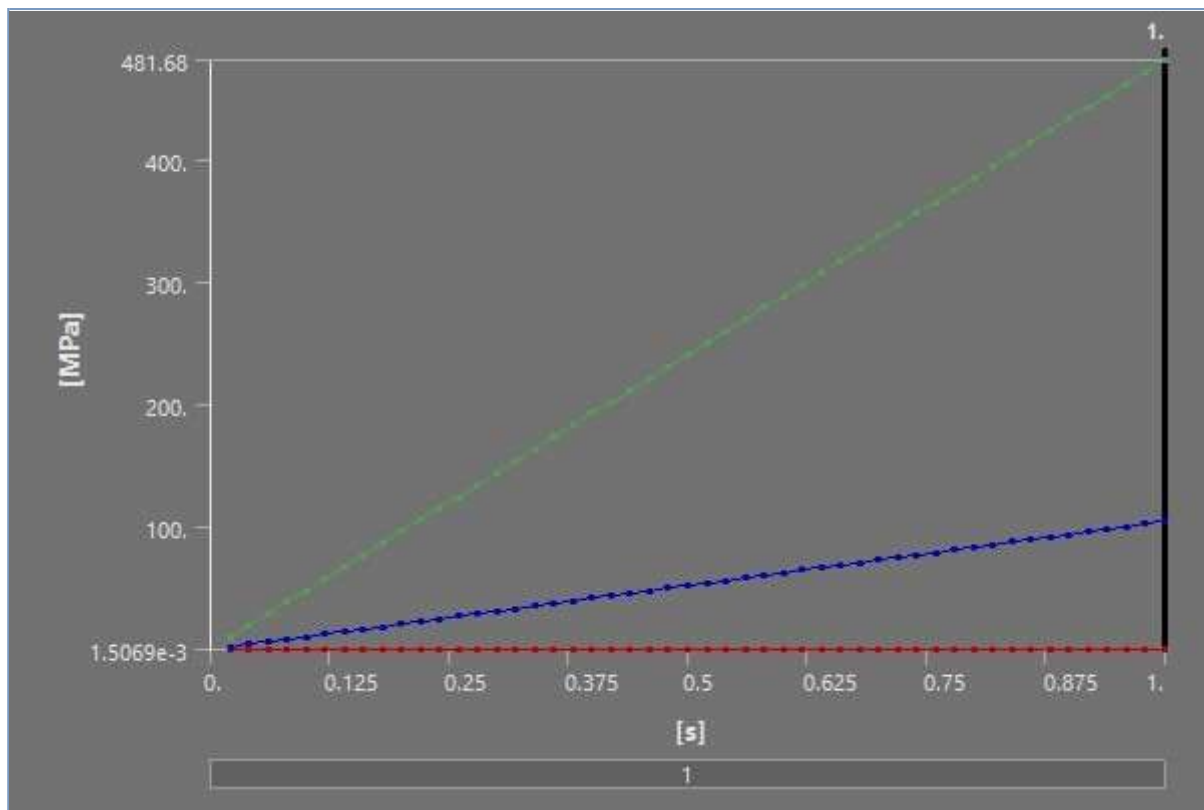


TABLE 21
Model (A4) > Static Structural (A5) > Solution (A6) > Equivalent Stress

Time [s]	Minimum [MPa]	Maximum [MPa]	Average [MPa]
2.e-002	1.5069e-003	9.6337	2.0909
4.e-002	3.0139e-003	19.267	4.1819
6.e-002	4.5208e-003	28.901	6.2728
8.e-002	6.0278e-003	38.535	8.3637
0.1	7.5347e-003	48.168	10.455
0.12	9.0417e-003	57.802	12.546
0.14	1.0549e-002	67.436	14.637
0.16	1.2056e-002	77.069	16.727
0.18	1.3563e-002	86.703	18.818
0.2	1.5069e-002	96.337	20.909
0.22	1.6576e-002	105.97	23.
0.24	1.8083e-002	115.6	25.091
0.26	1.959e-002	125.24	27.182
0.28	2.1097e-002	134.87	29.273
0.3	2.2604e-002	144.5	31.364
0.32	2.4111e-002	154.14	33.455
0.34	2.5618e-002	163.77	35.546
0.36	2.7125e-002	173.41	37.637
0.38	2.8632e-002	183.04	39.728
0.4	3.0139e-002	192.67	41.819
0.42	3.1646e-002	202.31	43.91
0.44	3.3153e-002	211.94	46.001
0.46	3.466e-002	221.57	48.091
0.48	3.6167e-002	231.21	50.182
0.5	3.7674e-002	240.84	52.273

0.52	3.9181e-002	250.48	54.364
0.54	4.0688e-002	260.11	56.455
0.56	4.2195e-002	269.74	58.546
0.58	4.3701e-002	279.38	60.637
0.6	4.5208e-002	289.01	62.728
0.62	4.6715e-002	298.64	64.819
0.64	4.8222e-002	308.28	66.91
0.66	4.9729e-002	317.91	69.001
0.68	5.1236e-002	327.54	71.092
0.7	5.2743e-002	337.18	73.183
0.72	5.425e-002	346.81	75.274
0.74	5.5757e-002	356.45	77.364
0.76	5.7264e-002	366.08	79.455
0.78	5.8771e-002	375.71	81.546
0.8	6.0278e-002	385.35	83.637
0.82	6.1785e-002	394.98	85.728
0.84	6.3292e-002	404.61	87.819
0.86	6.4799e-002	414.25	89.91
0.88	6.6306e-002	423.88	92.001
0.9	6.7813e-002	433.51	94.092
0.92	6.932e-002	443.15	96.183
0.94	7.0827e-002	452.78	98.274
0.96	7.2333e-002	462.42	100.36
0.98	7.384e-002	472.05	102.46
1.	7.5347e-002	481.68	104.55

FIGURE 4
Model (A4) > Static Structural (A5) > Solution (A6) > Shear Stress

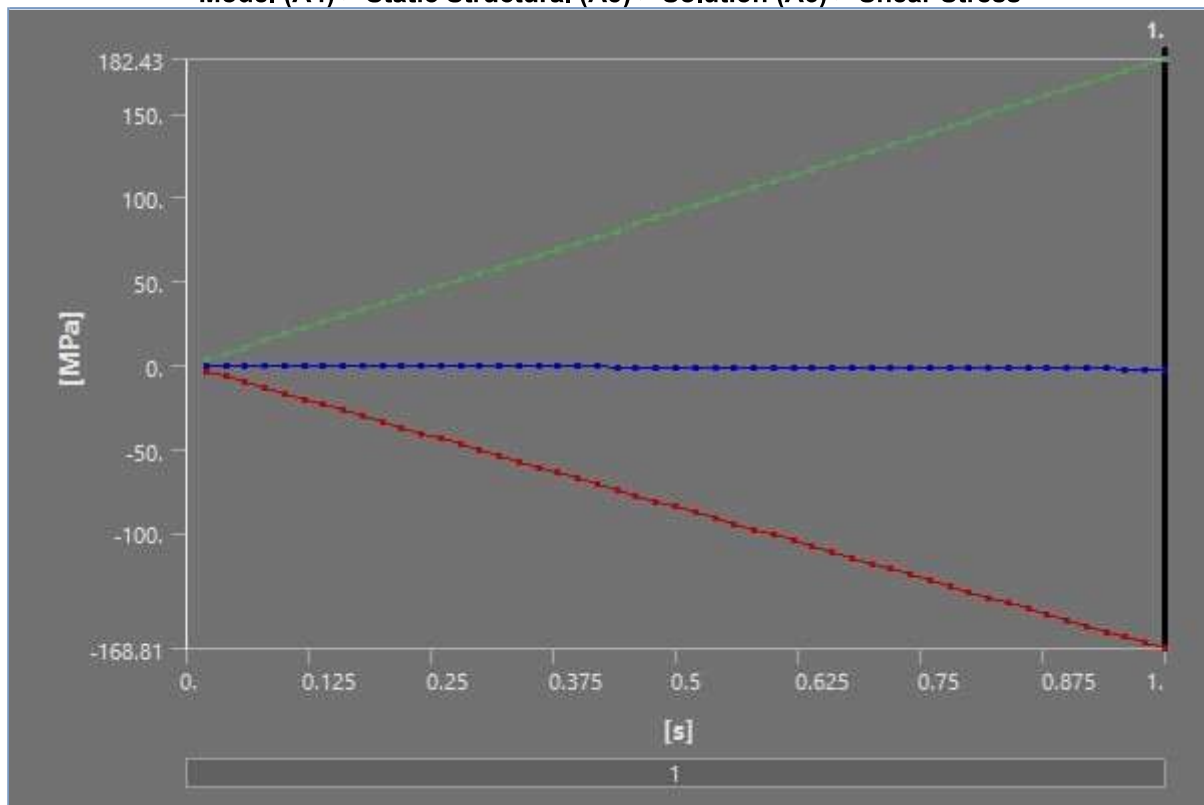


TABLE 22
Model (A4) > Static Structural (A5) > Solution (A6) > Shear Stress

Time [s]	Minimum [MPa]	Maximum [MPa]	Average [MPa]
2.e-002	-3.3761	3.6485	-4.4908e-002
4.e-002	-6.7523	7.297	-8.9816e-002
6.e-002	-10.128	10.946	-0.13472
8.e-002	-13.505	14.594	-0.17963
0.1	-16.881	18.243	-0.22454
0.12	-20.257	21.891	-0.26945
0.14	-23.633	25.54	-0.31436
0.16	-27.009	29.188	-0.35927
0.18	-30.385	32.837	-0.40417
0.2	-33.761	36.485	-0.44908
0.22	-37.137	40.134	-0.49399
0.24	-40.514	43.782	-0.5389
0.26	-43.89	47.431	-0.58381
0.28	-47.266	51.079	-0.62872
0.3	-50.642	54.728	-0.67362
0.32	-54.018	58.376	-0.71853
0.34	-57.394	62.025	-0.76344
0.36	-60.77	65.673	-0.80835
0.38	-64.146	69.322	-0.85326
0.4	-67.523	72.97	-0.89816
0.42	-70.899	76.619	-0.94307
0.44	-74.275	80.267	-0.98798
0.46	-77.651	83.916	-1.0329
0.48	-81.027	87.564	-1.0778
0.5	-84.403	91.213	-1.1227
0.52	-87.779	94.861	-1.1676
0.54	-91.155	98.51	-1.2125
0.56	-94.532	102.16	-1.2574
0.58	-97.908	105.81	-1.3023
0.6	-101.28	109.46	-1.3472
0.62	-104.66	113.1	-1.3922
0.64	-108.04	116.75	-1.4371
0.66	-111.41	120.4	-1.482
0.68	-114.79	124.05	-1.5269
0.7	-118.16	127.7	-1.5718
0.72	-121.54	131.35	-1.6167
0.74	-124.92	134.99	-1.6616
0.76	-128.29	138.64	-1.7065
0.78	-131.67	142.29	-1.7514
0.8	-135.05	145.94	-1.7963
0.82	-138.42	149.59	-1.8412
0.84	-141.8	153.24	-1.8861
0.86	-145.17	156.89	-1.9311
0.88	-148.55	160.53	-1.976
0.9	-151.93	164.18	-2.0209
0.92	-155.3	167.83	-2.0658
0.94	-158.68	171.48	-2.1107
0.96	-162.05	175.13	-2.1556
0.98	-165.43	178.78	-2.2005
1.	-168.81	182.43	-2.2454

TABLE 23
Model (A4) > Static Structural (A5) > Solution (A6) > Fracture Tool

Object Name	<i>Fracture Tool</i>
State	Solved
Scope	
Scoping Method	Crack Selection
Crack Selection Mode	Single Crack
Crack Selection	Pre-Meshed Crack
Definition	
Suppressed	No

TABLE 24
Model (A4) > Static Structural (A5) > Solution (A6) > Fracture Tool > Results

Object Name	SIFS (K1)	SIFS (K2)	SIFS (K3)
State	Solved		
Definition			
Type	SIFS		
Subtype	K1	K2	K3
Contour Start	1		
Contour End	6		
Active Contour	Last		
By	Result Set		
Set Number	Last		
Separate Data by Entity	No		
Suppressed	No		
Results			
Minimum	354.87 MPa·mm^(0.5)	-15.012 MPa·mm^(0.5)	-2.7319 MPa·mm^(0.5)
Maximum	366.21 MPa·mm^(0.5)	-13.503 MPa·mm^(0.5)	3.275 MPa·mm^(0.5)
Minimum Occurs On	SYS\Solid		
Maximum Occurs On	SYS\Solid		
Information			
Time	1. s		
Load Step	1		
Substep	50		
Iteration Number	50		

FIGURE 5
Model (A4) > Static Structural (A5) > Solution (A6) > Fracture Tool > SIFS (K1)

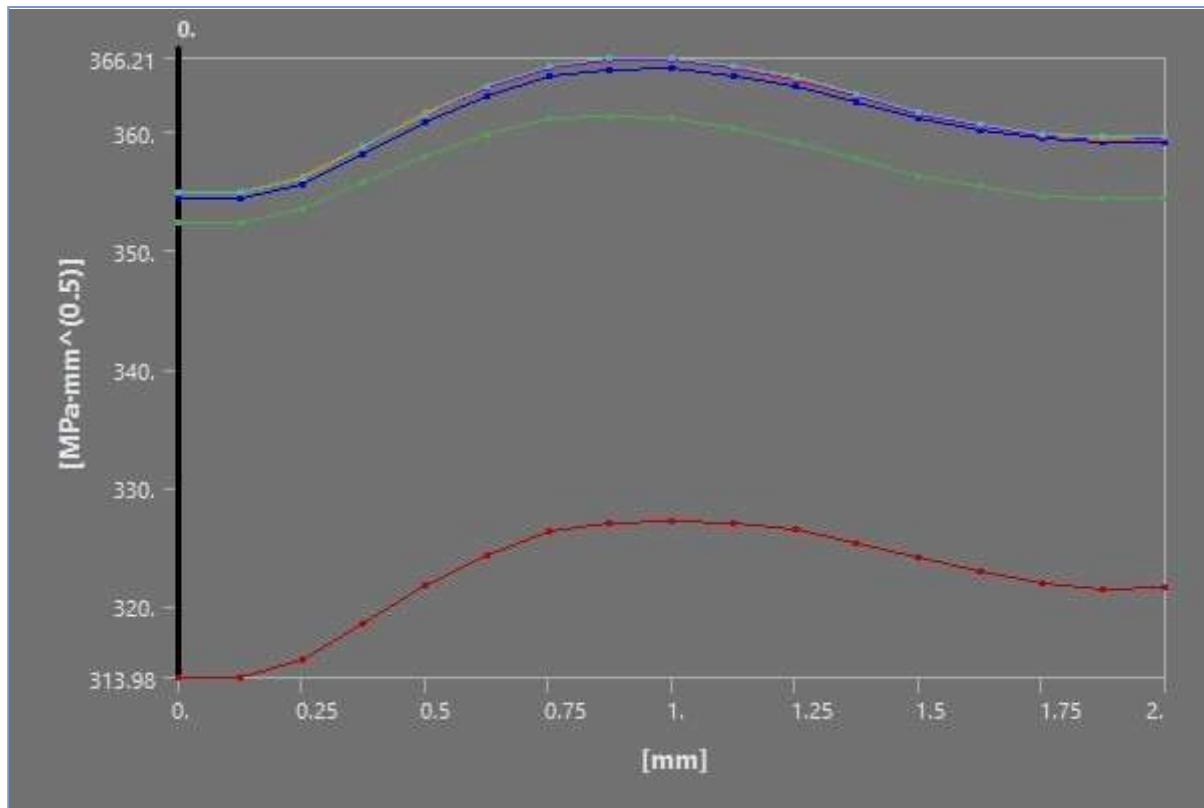


TABLE 25
Model (A4) > Static Structural (A5) > Solution (A6) > Fracture Tool > SIFS (K1)

Length [mm]	SIFS (K1) Contour 1 [MPa·mm ^{^(0.5)}]	SIFS (K1) Contour 2 [MPa·mm ^{^(0.5)}]	SIFS (K1) Contour 3 [MPa·mm ^{^(0.5)}]	SIFS (K1) Contour 4 [MPa·mm ^{^(0.5)}]	SIFS (K1) Contour 5 [MPa·mm ^{^(0.5)}]	SIFS (K1) Contour 6 [MPa·mm ^{^(0.5)}]
0.	314.01	352.38	354.39	354.84	354.95	354.9
0.125	313.98	352.35	354.35	354.81	354.91	354.87
0.25	315.51	353.45	355.6	356.07	356.19	356.15
0.375	318.59	355.68	358.17	358.69	358.81	358.78
0.5	321.77	357.98	360.86	361.43	361.56	361.55
0.625	324.28	359.77	363.03	363.67	363.81	363.81
0.75	326.24	361.09	364.71	365.41	365.56	365.58
0.875	327.	361.35	365.27	366.01	366.14	366.18
1.	327.21	361.09	365.29	366.02	366.15	366.21
1.125	326.97	360.27	364.75	365.38	365.52	365.59
1.25	326.41	359.17	363.88	364.39	364.53	364.61
1.375	325.36	357.73	362.57	362.94	363.09	363.17
1.5	324.17	356.32	361.21	361.49	361.63	361.71
1.625	322.99	355.34	360.17	360.44	360.56	360.63
1.75	321.95	354.63	359.36	359.66	359.77	359.82
1.875	321.48	354.41	359.09	359.42	359.52	359.56
2.	321.51	354.46	359.14	359.47	359.57	359.61

FIGURE 6
Model (A4) > Static Structural (A5) > Solution (A6) > Fracture Tool > SIFS (K2)

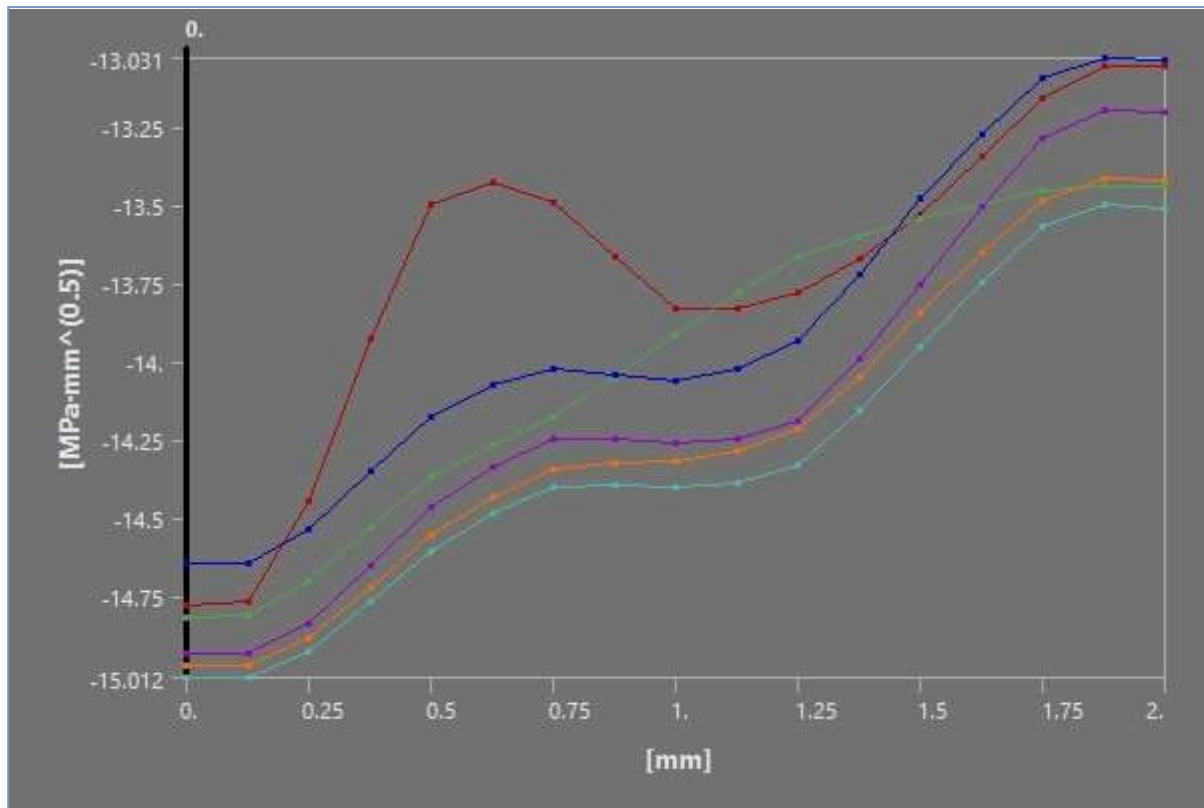


TABLE 26
Model (A4) > Static Structural (A5) > Solution (A6) > Fracture Tool > SIFS (K2)

Length [mm]	SIFS (K2) Contour 1 [MPa·mm ^{^(0.5)}]	SIFS (K2) Contour 2 [MPa·mm ^{^(0.5)}]	SIFS (K2) Contour 3 [MPa·mm ^{^(0.5)}]	SIFS (K2) Contour 4 [MPa·mm ^{^(0.5)}]	SIFS (K2) Contour 5 [MPa·mm ^{^(0.5)}]	SIFS (K2) Contour 6 [MPa·mm ^{^(0.5)}]
0.	-14.781	-14.817	-14.646	-14.935	-14.971	-15.011
0.125	-14.77	-14.813	-14.645			-15.012
0.25	-14.45	-14.704	-14.54	-14.836	-14.884	-14.927
0.375	-13.931	-14.533	-14.354	-14.652	-14.721	-14.767
0.5	-13.497	-14.371	-14.177	-14.468	-14.557	-14.607
0.625	-13.427	-14.27	-14.079	-14.339	-14.435	-14.488
0.75	-13.493	-14.176	-14.024	-14.251	-14.346	-14.404
0.875	-13.666	-14.047	-14.042	-14.249	-14.326	-14.397
1.	-13.829	-13.913	-14.064	-14.264	-14.319	-14.405
1.125	-13.836	-13.783	-14.025	-14.252	-14.288	-14.391
1.25	-13.779	-13.669	-13.932	-14.191	-14.22	-14.336
1.375	-13.67	-13.599	-13.721	-13.994	-14.047	-14.161
1.5	-13.528	-13.545	-13.483	-13.754	-13.846	-13.953
1.625	-13.342	-13.496	-13.272	-13.508	-13.652	-13.749
1.75	-13.158	-13.456	-13.097	-13.289	-13.484	-13.572
1.875	-13.059	-13.44	-13.031	-13.198	-13.418	-13.503
2.		-13.442	-13.039	-13.206	-13.424	-13.51

FIGURE 7
Model (A4) > Static Structural (A5) > Solution (A6) > Fracture Tool > SIFS (K3)

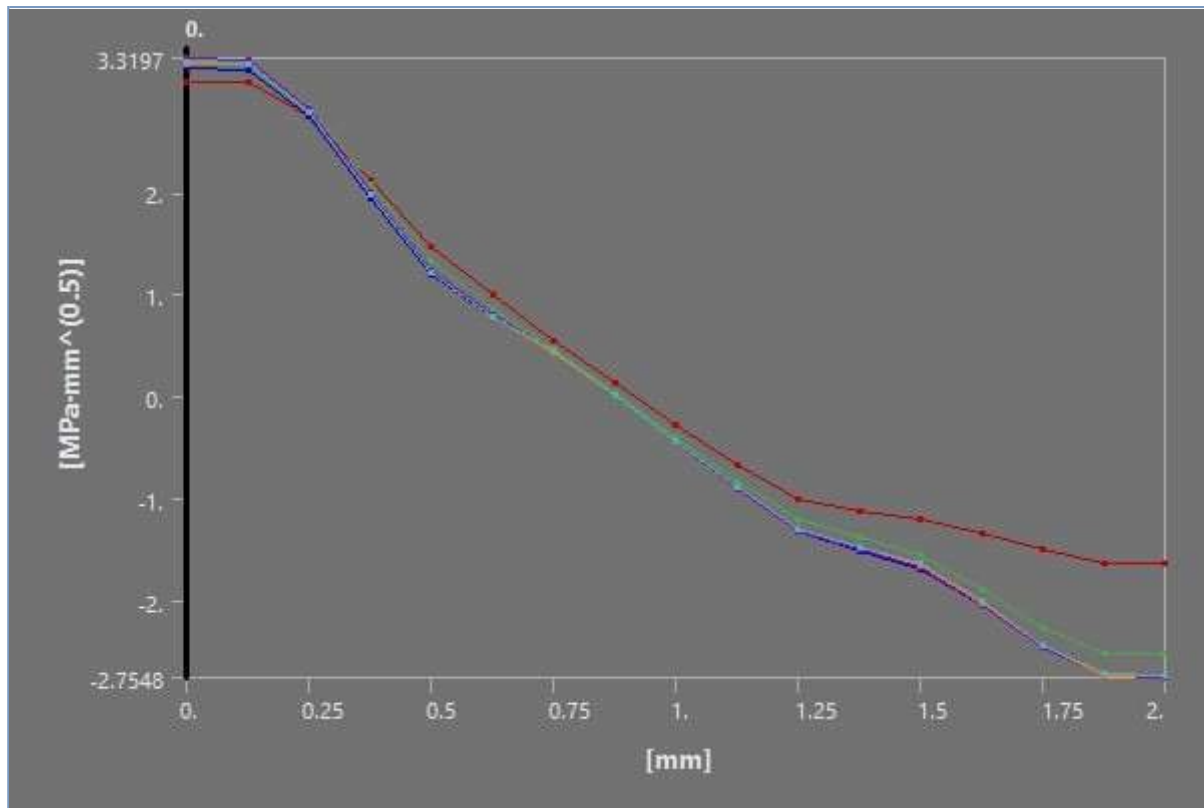


TABLE 27
Model (A4) > Static Structural (A5) > Solution (A6) > Fracture Tool > SIFS (K3)

Length [mm]	SIFS (K3) Contour 1 [MPa·mm ^{^(0.5)}]	SIFS (K3) Contour 2 [MPa·mm ^{^(0.5)}]	SIFS (K3) Contour 3 [MPa·mm ^{^(0.5)}]	SIFS (K3) Contour 4 [MPa·mm ^{^(0.5)}]	SIFS (K3) Contour 5 [MPa·mm ^{^(0.5)}]	SIFS (K3) Contour 6 [MPa·mm ^{^(0.5)}]
0.	3.0852	3.2959	3.2167	3.3197	3.2692	3.275
0.125	3.087	3.2901	3.2073	3.3094	3.2586	3.2642
0.25	2.7509	2.8508	2.7436	2.8276	2.7828	2.7869
0.375	2.1193	2.0791	1.9508	2.0077	1.9753	1.9781
0.5	1.4764	1.3322	1.2014	1.2345	1.2144	1.2163
0.625	0.99259	0.86303	0.77954	0.80176	0.7895	0.79144
0.75	0.55653	0.46474	0.43084	0.44572	0.43892	0.44114
0.875	0.13286	3.7866e-002	1.2684e-002	1.8815e-002	1.3251e-002	1.5604e-002
1.	-0.27681	-0.39013	-0.42591	-0.42676	-0.4303	-0.4274
1.125	-0.6783	-0.82519	-0.89896	-0.90078	-0.89495	-0.89
1.25	-1.0122	-1.2045	-1.3177	-1.3177	-1.3013	-1.2942
1.375	-1.1311	-1.3829	-1.5089	-1.5031	-1.4786	-1.4708
1.5	-1.2035	-1.5502	-1.6871	-1.6763	-1.6473	-1.6392
1.625	-1.3404	-1.8876	-2.0536	-2.0447	-2.021	-2.013
1.75	-1.4981	-2.2626	-2.4627	-2.4573	-2.4416	-2.4335
1.875	-1.6259	-2.5153	-2.7403	-2.7367	-2.726	-2.7172
2.	-1.6366	-2.5282	-2.7548	-2.7513	-2.7408	-2.7319

Material Data

Structural Steel

TABLE 28
Structural Steel > Constants

Density	7.85e-006 kg mm ⁻³
Coefficient of Thermal Expansion	1.2e-005 K ⁻¹
Specific Heat	4.34e+005 mJ kg ⁻¹ K ⁻¹
Thermal Conductivity	6.05e-002 W mm ⁻¹ K ⁻¹
Resistivity	1.7e-004 ohm mm

TABLE 29
Structural Steel > Color

Red	Green	Blue
132	139	179

TABLE 30
Structural Steel > Compressive Ultimate Strength

Compressive Ultimate Strength MPa
0

TABLE 31
Structural Steel > Compressive Yield Strength

Compressive Yield Strength MPa
250

TABLE 32
Structural Steel > Tensile Yield Strength

Tensile Yield Strength MPa
250

TABLE 33
Structural Steel > Tensile Ultimate Strength

Tensile Ultimate Strength MPa
460

TABLE 34
Structural Steel > Isotropic Secant Coefficient of Thermal Expansion

Zero-Thermal-Strain Reference Temperature K
295.15

TABLE 35
Structural Steel > S-N Curve

Alternating Stress MPa	Cycles	Mean Stress MPa
3999	10	0
2827	20	0
1896	50	0
1413	100	0
1069	200	0
441	2000	0
262	10000	0
214	20000	0
138	1.e+005	0
114	2.e+005	0
86.2	1.e+006	0

TABLE 36

Structural Steel > Strain-Life Parameters

Strength Coefficient MPa	Strength Exponent	Ductility Coefficient	Ductility Exponent	Cyclic Strength Coefficient MPa	Cyclic Strain Hardening Exponent
920	-0.106	0.213	-0.47	1000	0.2

TABLE 37**Structural Steel > Isotropic Elasticity**

Young's Modulus MPa	Poisson's Ratio	Bulk Modulus MPa	Shear Modulus MPa	Temperature K
2.e+005	0.3	1.6667e+005	76923	

TABLE 38**Structural Steel > Isotropic Relative Permeability**

Relative Permeability
10000